Junyan Tan (Samuel)

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Research Interests

Population Genetics; Quantitative Genetics; Machine Learning.

Research Experience

- **University of Michigan, Department of Statistics** *May 2025 – Jan 2026*
  PI: Prof. Jonathan Terhorst Ann Arbor, USA
  - Co-created `demestats`, a Python library for demographic inference; implemented a JAX-based optimizer that automates demographic parameter estimation (replacing manual visual calibration in earlier workflows). Co-author on the demestats paper.
  - Applied demestats to real datasets, including an internal unphased Michigan snake dataset, and participated in the Genomic History Inference Strategies Tournament (GHIST) using demestats for inferring demographic parameters.
- **University of Michigan School of Public Health, Department of Biostatistics** *Aug 2024 – May 2025*
  PI: Prof. Irina Gaynanova Ann Arbor, USA
  - Developed a zero-shot LLM framework `Gluco-LLM` with Retrieval-Augmented Generation (RAG) for CGM time-series forecasting, achieving up to a 30% reduction in error and 50% lower compute time compared to strong time-series baselines.
  - Maintained and extended the `iglu` R package for CGM analysis (bug fixes, new features, tests, documentation) and enhanced the Shiny-based visualization interface.
  - Curated and processed `Awesome-CGM`, a widely used public CGM dataset for 1,000+ individuals with over 200 stars on Github. Coordinated analyses for a multi-institution consensus study with five external CGM software teams.
- **Academy of Mathematics and Systems Science, Chinese Academy of Sciences** *May 2024 – Aug 2024*
  PI: Prof. Qizhai Li Beijing, China
  - Developed a novel test statistic for multimodal regression with an R implementation and theoretical guarantees.
  - Demonstrated 1.5–2× higher statistical power on simulated settings compared to prior tests; prepared an internal report.

Education

- **University of Southern California** *Current*
  Ph.D. Computational Biology and Bioinformatics Los Angeles, USA
- **University of Michigan** *2026*
  M.S. Biostatistics Ann Arbor, USA
- **University of Michigan** *2025*
  B.S. Data Science & Statistics, Honors in Data Science Ann Arbor, USA

Honors and Awards

- **Phi Beta Kappa** *Apr 2025*
  Phi Beta Kappa - Alpha of Michigan Chapter
- **Best Poster Award** *Mar 2025*
  Michigan Student Symposium for Interdisciplinary Statistical Sciences

Publications, Preprints & Selected Works

[1] Dilber, E., Liang, J., **Tan, J.**, Terhorst, J. (2026). *Faster inference of complex demographic models from large allele frequency spectra. Bioinformatics*, btag511.

[2] **Tan, J.** (2025). *Evaluating the Zero-Shot Predictive Ability of Large Language Models for Continuous Glucose Monitoring Data.* Undergraduate Honors Thesis, University of Michigan.

[3] Xu, X., Kok, N., **Tan, J.**, et al. (2024). *Awesome-CGM: List of CGM datasets.* Version 2.0.0. GitHub.

Talks & Presentations

- **Evaluating the Zero-Shot Predictive Ability of LLMs for CGM Time Series** *Mar 2025*
  MSSISS 2025 — Poster (Best Poster Award)
  - Presented honors-thesis results on zero-shot LLM forecasting for CGM time series; highlighted gains and the role of retrieval-augmented generation (RAG).

Skills

- **Programming Languages:** R, Python, C++, C, SQL, Java, JavaScript
- **Machine Learning & Stats:** LLM, CNN, MCMC, HMM, Time-Series Forecasting
- **Tools & Frameworks:** JAX, PyTorch, TensorFlow, Hugging Face, Git, GCP, HPC Computing
- **Languages:** English, Chinese, Japanese (fluent); French (fluent); Spanish (working knowledge)

Mentorship & Outreach

- **Data Science Consultant** *Jan 2024 – Apr 2024*
  Industrial Sewing and Innovation Center (ISAIC)
  - Analytics engagement for a Detroit-based nonprofit: Python-based analyses and visualizations (seaborn, matplotlib) and a concise report used in grant applications.
  - *My contribution:* designed the analysis plan, created the visuals, and co-authored the report.
- **Education Committee** *Jan 2024 – Aug 2024*
  Michigan Data Science Team (MDST)
  - Programmed training content for student practitioners, emphasizing statistical foundations and theory in data science (regularization, cross-validation, Bayesian methods).
  - *My contribution:* led curriculum design and delivery and mentored the project team.